

Electromagnetism 2

(spring semester 2026)

Lecture 5

Magnetic dipoles; EM induction

- ❖ Magnetic dipoles and their properties
- ❖ Faraday's law of electromagnetic induction
- ❖ Electromotive force in a loop of wire

Previous lecture

- ❖ Electrostatic potential of a point-like electric dipole:

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

- ❖ Electric dipoles in uniform fields: zero net force; non-zero torque; dipole moments tend to align along the field lines.

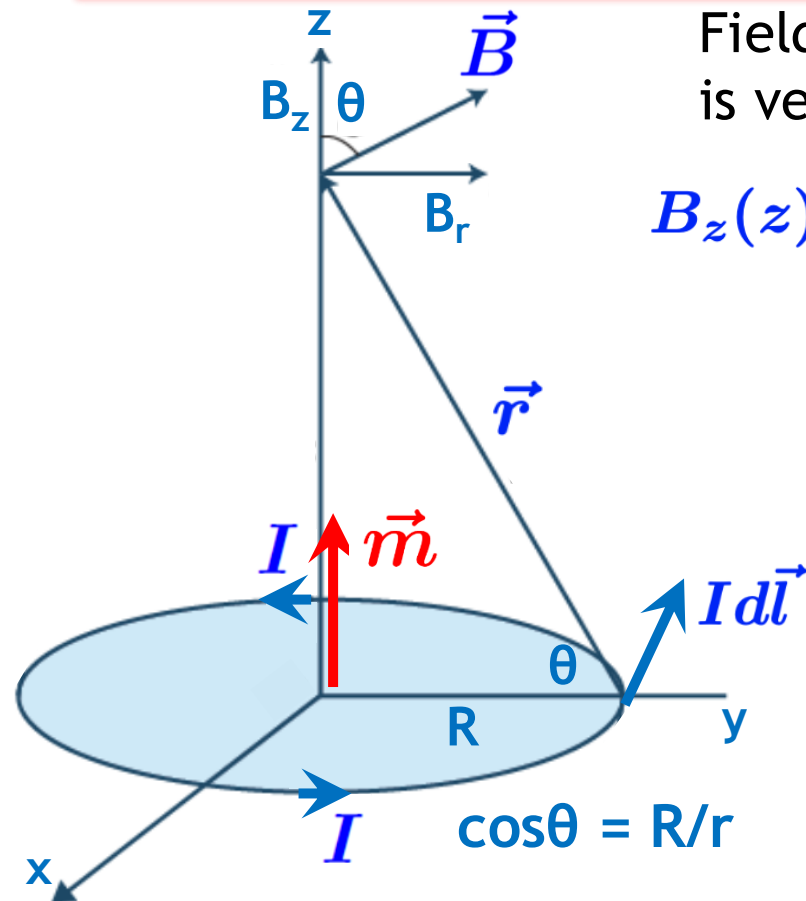
$$\vec{\tau} = \vec{p} \times \vec{E} \quad U = -\vec{p} \cdot \vec{E}$$

- ❖ Electric dipoles in non-uniform fields: non-zero net force.
A dipole aligned with the field is attracted towards stronger field.

$$\vec{F} = (\vec{p} \cdot \nabla) \vec{E}$$

- ❖ Electrically neutral objects with non-zero dipole moments ($\mathbf{Q=0}$, $\mathbf{p \neq 0}$), e.g. water molecules, experience electrostatic forces.

Magnetic dipole



Field of a circular current loop on the z axis is vertical by symmetry,

$$B_z(z) = \frac{\mu_0}{4\pi} \oint_L \frac{Idl}{r^2} \cos \theta = \frac{\mu_0 I \cos \theta \cdot 2\pi R}{4\pi r^2}$$

$$= \frac{\mu_0 I \cdot \pi R^2}{2\pi r^3} = \frac{\mu_0 m}{2\pi r^3} \xrightarrow{|z| \gg R} \frac{\mu_0 m}{2\pi |z|^3}$$

Identical to the electric dipole field,

$$E_z(z) = \frac{p}{2\pi\epsilon_0 |z|^3} \quad (\text{lecture 4})$$

with a substitution $\mathbf{p} \rightarrow \mathbf{m}$, $1/\epsilon_0 \rightarrow \mu_0$.

More generally, fields of point-like magnetic and electric dipoles are identical at all points in space.

Magnetic dipole moment:

$$\vec{m} = I\vec{S}$$

A planar *current loop* is equivalent to a *pair of magnetic charges*.

A current loop, an electron or a hydrogen atom are magnets.

Dipole in a uniform field

A square current loop
(area $S=a \times a$) in a
uniform external $\mathbf{B}$ field

No net force acting on the loop:

$$\vec{F} = \oint_L I d\vec{l} \times \vec{B} = -I \vec{B} \times \oint_L d\vec{l} = \vec{0}$$

Torque exerted on a loop:

$$\begin{aligned} \tau &= F a \sin \theta = I B a \cdot a \sin \theta \\ &= I S B \sin \theta \end{aligned}$$

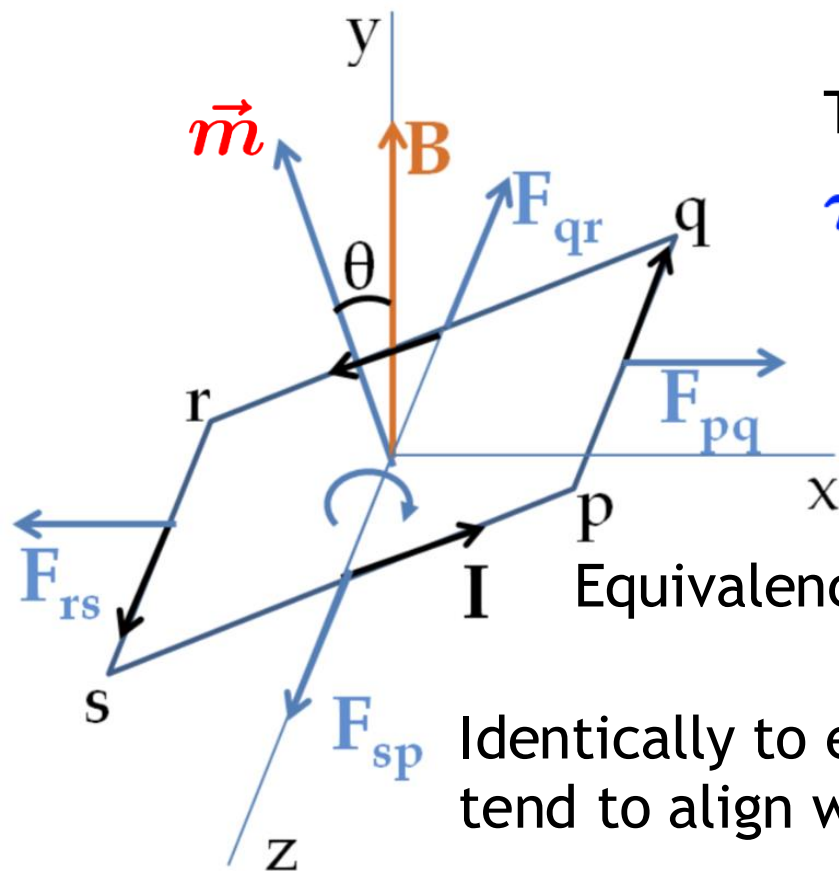
Using the magnetic dipole moment,
in the vector form, $\vec{\tau} = \vec{m} \times \vec{B}$

Equivalence to electric dipoles: $\vec{\tau} = \vec{p} \times \vec{E}$
(lecture 4)

Identically to electric dipoles, magnetic dipoles
tend to align with the field; the potential energy is

$$U = -\vec{m} \cdot \vec{B}$$

Example: a compass needle.



Dipole in a non-uniform field

The analogy extends to *non-uniform* external fields.

Force acting on an electric dipole:

(lecture 4)

$$\vec{F} = (\vec{p} \nabla) \vec{E}$$

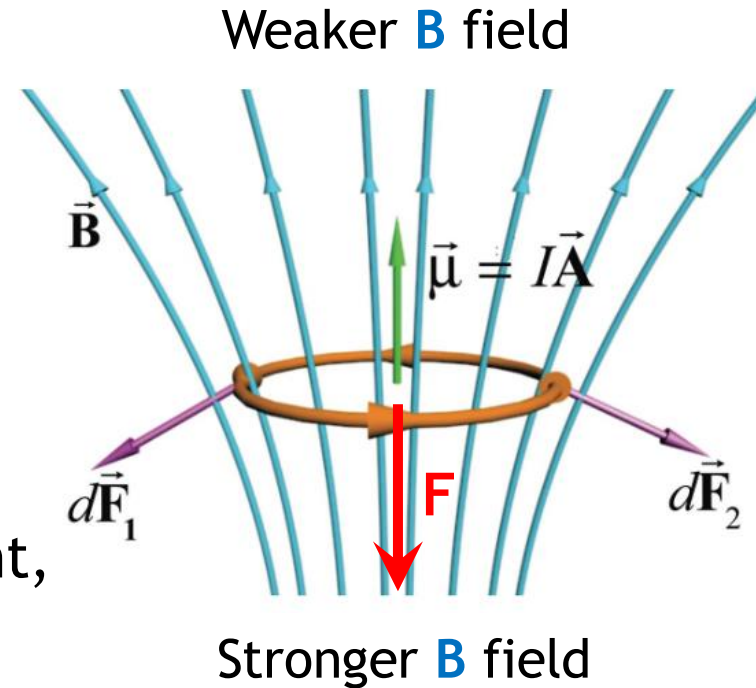
Analogously, force on a magnetic dipole:

$$\vec{F} = (\vec{m} \nabla) \vec{B}$$

In case both the dipole moment and the magnetic field only have the **x** component,

$$F_x = m_x \frac{\partial B_x}{\partial x}$$

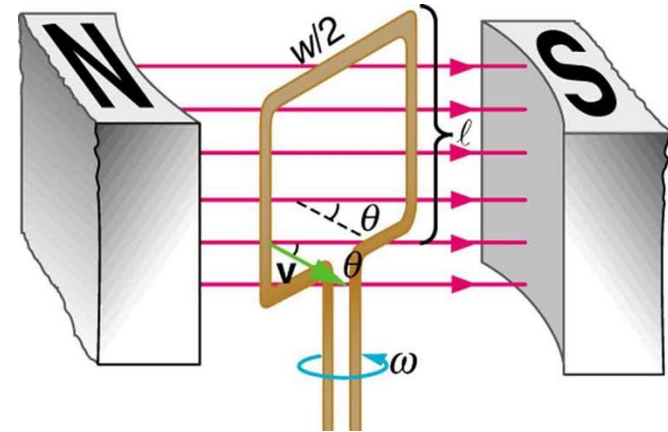
- ❖ A dipole tends to align with the external field.
- ❖ For a dipole aligned with the field, the force is directed towards the region of stronger field.
- ❖ This leads to attraction of parallel electric dipoles and parallel magnetic dipoles (including permanent magnets).



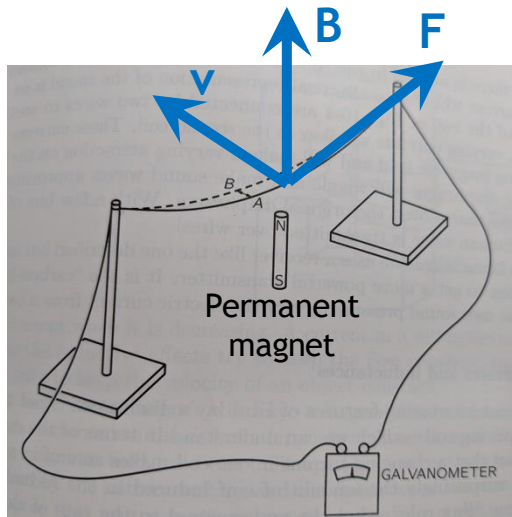
Induced currents

Electric currents, and magnets, produce magnetic fields

- ❖ B-field produces Lorentz force on moving charges: $\vec{F} = q\vec{v} \times \vec{B}$
- ❖ Applications: AC generators and motors; measurements of currents and B-fields.
- ❖ There are magnetic forces between currents.



Do magnets produce electric fields or currents?



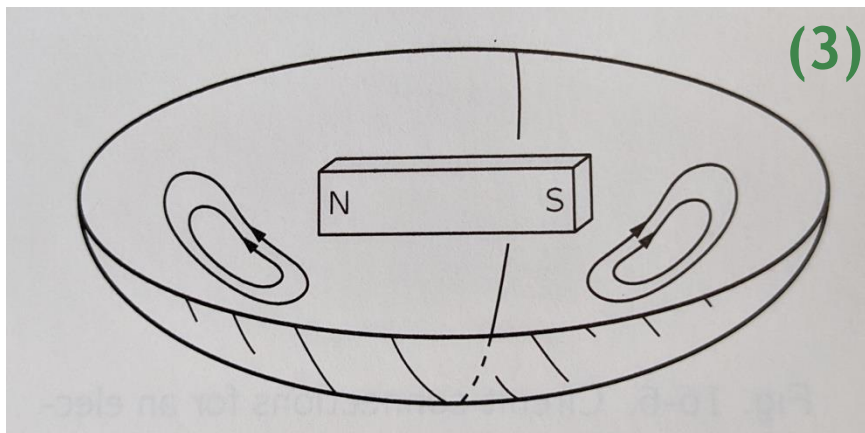
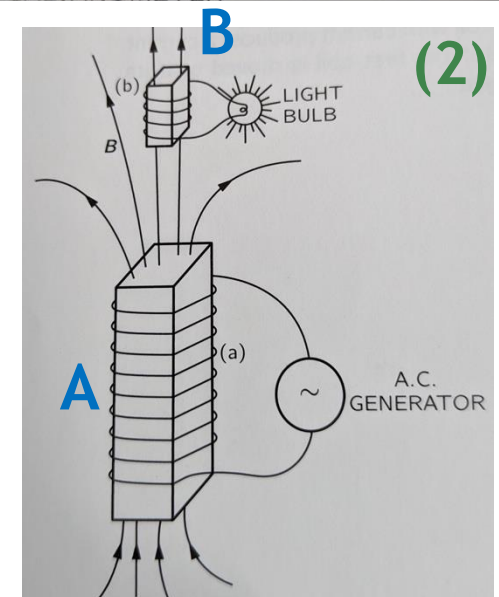
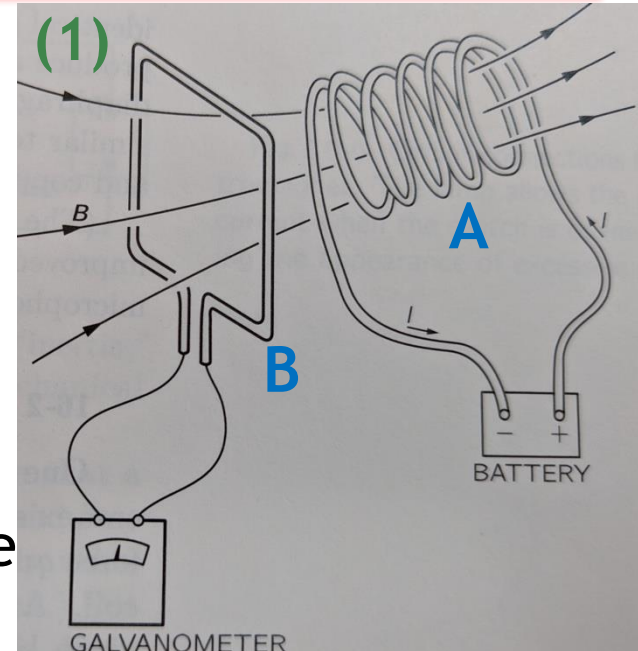
- ❖ Transverse movement of the *wire*: a pulse of current observed, due to the Lorentz force.
- ❖ Transverse movement of the *magnet*: no Lorentz force but *a pulse of current is still observed*.
- ❖ This is a new phenomenon: *induced current* due to the variation of the magnetic field.

Examples of induced currents

(1) Replacing a permanent magnet with a coil: a pulse of current is observed in loop **B** due to *change of current, position or orientation* of the coil **A**.

(2) Alternating current is induced in coil **B** by an alternating current in coil **A**.

(3) Magnetic levitation: a magnet placed above superconductor surface is repelled by the **B**-field created by the induced *eddy current*.



Faraday's law

Faraday's law for the *induced electric field*, in the differential form:

(universally valid)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Unlike the electrostatic field (*lecture 1*), the **electric field** is **not conservative** in the general case

Using the curl theorem, for *any* loop **L** and any surface enclosed,

$$\oint_L \vec{E} d\vec{l} = \int_S (\nabla \times \vec{E}) d\vec{S} = - \int_S \frac{\partial \vec{B}}{\partial t} d\vec{S} \quad (\text{lecture 2})$$

Faraday's law in the integral form:

$$\underbrace{\oint_L \vec{E} d\vec{l}}_{\text{Line integral of the induced electric field around a closed curve}} = - \underbrace{\int_S \frac{\partial \vec{B}}{\partial t} d\vec{S}}_{\text{Rate of change of magnetic flux through any surface enclosed by the curve}}$$

Lenz's law (the minus sign):

the direction of the induced current (if one were to flow) is such that its effect would oppose the change in magnetic flux.

Line integral of the induced electric field around a closed curve

Rate of change of magnetic flux through *any* surface enclosed by the curve

Electromotive force in a loop

- ❖ Electromotive force (*EMF*) in a circuit: $\mathcal{E} = \frac{1}{q} \oint_L \vec{F}_{\text{Lorentz}} d\vec{l}$
- ❖ Two ways of inducing EMF in a loop of wire, due to the Lorentz force acting on a charge q :

$$\vec{F}_{\text{Lorentz}} = q(\underbrace{\vec{E}}_{\text{by electromagnetic induction, i.e. by variation of the magnetic field (Faraday's law)}} + \underbrace{\vec{v} \times \vec{B}}_{\text{by motion of the wire (deformation or change of orientation, } \vec{v} \neq 0 \text{); known as } \textit{motional EMF}}})$$

- ❖ The two phenomena are distinct and independent. However, both lead to the same formula for the EMF:

$$\mathcal{E} = - \frac{d\Phi_B}{dt} \quad \text{(rate of change of the magnetic flux through the loop)}$$

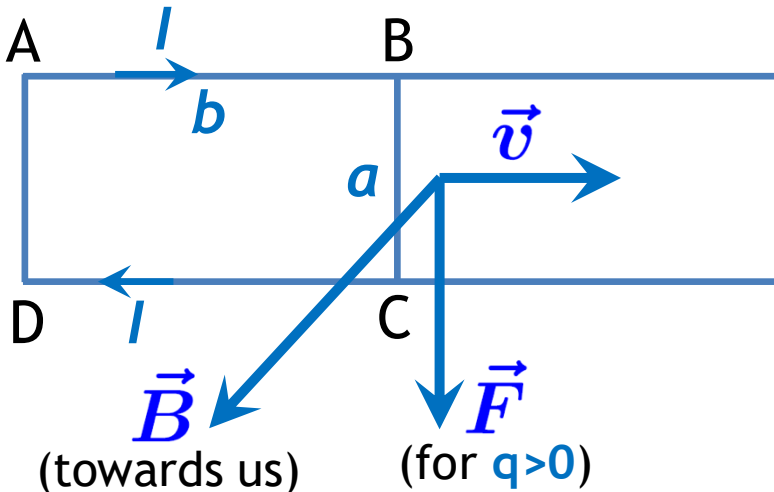
Proof of the EMF formula

Via electromagnetic induction:

The formula follows directly from Faraday's law,

$$\mathcal{E} = \frac{1}{q} \oint_L \vec{F}_{\text{Lorentz}} d\vec{l} = \frac{1}{q} \oint_L q \vec{E} d\vec{l} = \underbrace{\oint_L \vec{E} d\vec{l} = - \int_S \frac{\partial \vec{B}}{\partial t} d\vec{S}}_{\text{Faraday's law}}$$

Via deformation of the wire:



A fixed U-shaped loop;
a crossbar (BC) moving with a speed $\mathbf{v}$;
uniform static external $\mathbf{B}$ -field.

Lorentz force: $\vec{F} = q\vec{v} \times \vec{B}$

Motional EMF:

$$|\mathcal{E}| = \frac{1}{q} \int_B^C \vec{F} d\vec{l} = \frac{1}{q} \cdot qvBa = Ba \frac{db}{dt} = \frac{d(abB)}{dt} = \frac{d\Phi_B}{dt}$$

Example: dipole field and EMF

EMF in a loop of wire L due to varying magnetic field of a moving magnet:



For $L \gg r$, the point-like dipole approximation is valid.

Magnetic flux through the loop:

$$\Phi_B \approx \frac{\mu_0 m}{2\pi L^3} \cdot \pi r^2 = \frac{\mu_0 m r^2}{2L^3}$$

Magnitude of the electromotive force:

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = \frac{d\Phi_B}{dL} \cdot \frac{dL}{dt} = -\frac{3\mu_0 m r^2}{2L^4} \cdot (-v) = \frac{3\mu_0 m v r^2}{2L^4}$$

Summary

- ❖ Our current (still incomplete) understanding of the equations of electromagnetism *in free space*:

Gauss law (*universally valid*)

$$\nabla \vec{E} = \rho / \epsilon_0$$

Faraday's law (*universally valid*)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Absence of magnetic poles
(*universally valid*)

$$\nabla \vec{B} = 0$$

Ampere's law (*static field only*)

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

- ❖ Variation of magnetic field gives rise to electric field.
- ❖ Does variation of electric field give rise to magnetic field?